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SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: **B.E / B.Tech**

Course Code: **CSA5102**

Course Name: **Cryptography and Network Security**

Duration: **1: 30 Hr**

Date : 02/11/2026

Max Marks: **50 Marks**

45/50

CONTINUOUS INTERNAL ASSESSMENT 1

Course Outcomes

On successful completion of the course, students will be able to:

CO	Course Outcome Statement
CO1	Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for ensuring data security.
CO2	Analyze public-key cryptosystem strategies using RSA and key exchange for strengthening information security.

Answer all the Questions

1. A security analyst at a small firm is asked to demonstrate classical encryption to new trainees. The plaintext "MEETATNOON" must be encrypted using a Caesar cipher with shift 7. Show the encryption and decryption process step by step, first converting each letter to its numeric value (A=0, B=1, ..., Z=25), applying the formulas C = (P + 7) mod 26 and P = (C − 7) mod 26, and then converting the numbers back to letters. Next, encrypt the same plaintext with a Vigenère cipher using the repeating key "SUN", showing the key shift for every letter position. Compare the Caesar cipher with the Vigenère cipher in terms of
   (a) key space
   (b) treatment of repeated patterns such as the double letters EE and OO and the repeated letters T and N, and

(c) vulnerability to cryptanalysis

Recommend one of the two only for a constrained educational demonstration, and justify your choice. Finally, explain why neither cipher is suitable for protecting modern confidential data, considering key size, determinism, lack of integrity protection and the strength of modern algorithms such as AES-256-GCM. **(10)**

(CO1) (BL3)

2. A cryptography instructor wants to show how matrix algebra is used in classical block ciphers.

Encrypt the plaintext "MATH" using a 2 × 2 Hill cipher with key K = [[3, 2], [5, 7]], using A=0, B=1,
..., Z=25. Show the conversion of the plaintext into column vectors (MA and TH), and show the matrix multiplication and modulo-26 reduction for each block to obtain the ciphertext. Determine whether the key matrix is invertible modulo 26 by computing its determinant and checking gcd(det, 26) = 1. Find the modular inverse of the determinant, compute the inverse key matrix K^{-1} using the adjugate method, and verify that $K \cdot K^{-1} \equiv I \pmod{26}$. Explain how K^{-1} enables decryption by applying it to the ciphertext blocks and recovering "MATH". Show why the alternative key K = [[2, 3], [4, 5]] is unacceptable by calculating its determinant and its gcd with 26. Finally, identify and explain a major weakness of the Hill cipher, such as its linearity and vulnerability to known-plaintext attacks.

(10) (CO1) (BL4)

3. a) A hospital network of 200 clinics must send large MRI archives (about 1 GB each) to a central radiology server, and every clinic must also exchange encryption keys with the server securely. The IT team is choosing between a symmetric cipher (AES-256-GCM) and the RSA public-key algorithm. Compare the two in terms of

(a) key usage (who holds which key),
(b) performance and maximum data size per operation,
(c) scalability, including the number of keys needed if all 200 clinics must communicate pairwise versus using RSA key pairs, and key-distribution requirements, including the need for certificates to authenticate public keys. **(5) (CO1) (BL4)**

b) Using this comparison, design a hybrid encryption workflow that uses each algorithm where it is most appropriate, listing the steps carried out by the sending clinic (certificate validation, DEK and nonce generation, AES-GCM encryption with associated data, RSA-OAEP key wrapping, and packaging) and by the receiving server (key unwrapping, decryption, and tag verification). Also explain how secure key storage should be handled (HSM/KMS, non-exportable keys, role-based access, MFA, audit logging, backup and recovery), and how key rotation should be handled (per-file DEKs, periodic rotation of the key-encryption key, re-wrapping, retirement of old keys, and emergency rotation after suspected compromise).

(5) (CO2) (BL4)

4. For an RSA demonstration in a cryptography laboratory, let the primes be p = 11 and q = 19, with public exponent e = 7. Calculate the modulus n and Euler's totient φ(n), and confirm that gcd(e, φ(n)) = 1. Determine the private exponent d such that $e \cdot d \equiv 1 \pmod{\varphi(n)}$ using the extended Euclidean algorithm, showing every division step and the back-substitution. State the public key and the private key. Encrypt the message M = 5 using C = M^e mod n, showing the intermediate modular exponentiation steps. Verify the result by decrypting the ciphertext using M = C^d mod n (you may use the Chinese Remainder Theorem to simplify the calculation) and confirm that the original message is recovered. Finally, explain why real systems require much larger parameters (2048-bit or 3072-bit moduli)

and careful padding such as RSA-OAEP for encryption and RSA-PSS for signatures, discussing factoring attacks, determinism, malleability and small-message attacks on textbook RSA.
(10) (CO2) (BL4)

5. A financial institution transmits confidential customer information across a network and must protect it against unauthorized access. Apply suitable classical encryption techniques, such as Caesar cipher, Playfair cipher, and Vigenère cipher, to encrypt and decrypt the given plaintext. Extend your solution by applying an appropriate symmetric or asymmetric cipher to secure the same information. Demonstrate the encryption and decryption procedures, compare the techniques in terms of key usage and security, and justify which approach is more suitable for protecting sensitive financial data. **(10) (CO1) (BL3)**

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CONTINUOUS INTERNAL ASSESSMENT-1

1) **Caesar Cipher and Vigenere Cipher**

(a) **Caesar Cipher Encryption**

Given:-

* plaintext = MEETATNOON
* Shift key = 7
* Alphabet mapping A=0, B=1, ... Z=25

Encryption formula:-

$$C = (P + 7) \bmod 26$$

↳ Shift key

Decryption formula:-

$$P = (C - 7) \bmod 26$$

↳ shift key

Plaintext	M	E	E	T	A	T	N	O	O	N
P	12	4	4	19	0	19	13	14	14	13

Plain text	P	Calculation	C	Ciphertext
M	12	(12+7) mod 26	19	T
E	4	(4+7) mod 26	11	L
E	4	(4+7) mod 26	11	L
T	19	(4+7) mod 26	0	A

Plaintext	P	calculation	C	Cipher text
A	0	(0+7) mod 26	7	H
T	19	(19+7) mod 26	0	A
N	13	(13+7) mod 26	20	U
O	14	(14+7) mod 26	21	V
O	14	(14+7) mod 26	21	V
N	13	(13+7) mod 26	20	U

result of encryption:-

TLLAHAUVVU

(b) Caesar cipher Decryption

Ciphertext	C	Calculation	P	Recovered letter
T	19	(19-7) mod 26	12	M
L	11	(11-7) mod 26	4	E
L	11	(11-7) mod 26	4	E
A	0	(0-7) mod 26	19	T
H	7	(7-7) mod 26	0	A
A	0	(0-7) mod 26	19	T
U	20	(20-7) mod 26	13	N
V	21	(21-7) mod 26	14	O
V	21	(21-7) mod 26	14	O
U	20	(20-7) mod 26	13	N

Recovered plaintext after Decryption:-

MEETATNOON

(c) Vigenere Cipher Encryption

Given:-

* Plaintext :- MEETATNOON
* Key= SUN

The key is repeated until it matches the plaintext length

Plaintext	M	E	E	T	A	T	N	O	O	N
Repeated key	S	U	N	S	U	N	S	U	N	S
key value	18	20	13	18	20	13	18	20	13	18
Plaintext value	12	4	4	19	0	19	13	14	14	13
ciphertext	E	Y	R	L	U	G	F	I	B	F

Encryption formula:-

$$C_i = (P_i + K_i) \bmod 26$$

$$C_1 = (12+18) \bmod 26 = 4 \Rightarrow E$$

$$C_2 = (4+20) \bmod 26 \Rightarrow 24 \Rightarrow Y$$

Final Encryption result:-

EYRLUGFIBF

(D) Comparison

Parameter	Caesar	Vigenere
Key space	26 shifts	$26^3 = 17576$ keys
Encryption	Fixed shift	multiple shifts
Repeated patterns	Same letters produce Same output	Same letter may produce different output.
Vulnerability	Brute force and Frequency analysis	Kasiski and frequency analysis.

In caesar cipher, FE becomes LL and OO becomes VV.

In Vigenere, EE becomes YR and oo becomes IB because the key changes the shift.

(E) Recommendation

Caesar is recommended for educational demonstrations because it is simple to understand and implement. Vigenere is useful for demonstrating polyalphabetic encryption.

(F) Why Neither is suitable for modern security :-

* Small key space makes them vulnerable to attacks.
* Deterministic encryption reveals repeated patterns.
* Neither provides data integrity or authentication.
* Both are weaker than modern algorithms.

AES-256 GCM provides 256 bits encryption, confidentiality integrity and authentication through an authentication tag.

2) Hill cipher Using Matrix Algebra:-

Given:-

Plaintext = MATH

$$K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$$

(A) Plaintext conversion

A=0, B=1, ... Z=25

$$MA = \begin{bmatrix} 12 \\ 0 \end{bmatrix}, TH = \begin{bmatrix} 19 \\ 7 \end{bmatrix}$$

Encryption formula:-

$$C = KP \bmod 26$$

(B) Encryption of MA

$$C_1 = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 12 \\ 0 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 36 \\ 60 \end{bmatrix} \bmod 26 = \begin{bmatrix} 10 \\ 8 \end{bmatrix}$$

- 10 = K, 8 = I.

(C) Encryption of TH:-

$$C_2 = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 19 \\ 7 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 71 \\ 144 \end{bmatrix} \bmod 26 = \begin{bmatrix} 19 \\ 14 \end{bmatrix}$$

$19 = T$, $14 = O$.

(D) check Invertibility

$$|K| = (3 \times 7) - (2 \times 5)$$

$$= 21 - 10 = 11$$

$$\gcd(11, 26) = 1$$

Therefore, the matrix is invertible modulo 26.

(E) Modular Inverse:-

$$11^{-1} \bmod 26 = 19$$

Using adjoint

$$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \bmod 26$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

(F) Verification:-

$$KK^{-1} = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 27 & 52 \\ 78 & 105 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Hence, the inverse is correct.

(G) Decryption :-

$$P = K^{-1} C \bmod 26$$

For KI :-

$$\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 10 \\ 8 \end{bmatrix} \bmod 26 = \begin{bmatrix} 12 \\ 0 \end{bmatrix} = MA$$

For TO :-

$$\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 19 \\ 14 \end{bmatrix} \bmod 26 = \begin{bmatrix} 19 \\ 7 \end{bmatrix} = TH$$

(H) Alternative key :-

$$K = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

$$|K| = (2 \times 5) - (3 \times 4) = -2$$

$$\gcd(-2, 26) = 2 \neq 1$$

Therefore, this key is not ~~invertible~~ invertible and is unacceptable.

(I) Major weakness :-

The Hill cipher is linear. Attacks can use known plaintext ciphertext pairs to determine the key matrix. It also lacks authentication and Integrity protection.

3) (a). AES-256-GCM vs RSA

A hospital network has 200 clinics transferring MRI archeiving of approximately 1 GB each.

Comparison:-

Parameter	AES-256-GCM	RSA
Type	Symmetric	Asymmetric
keys	Same secrets key	Public and private key
Speed	Very fast	Comparatively slow
Data size	Suitable for large files	Suitable only for Small data
Security	Confidentiality and integrity	Public key encryption
key distribution	Secure Secret key Sharing	Certificate authenticate Public keys.

Key Usage:-

AES :-

* Sender and reciver share one secret key
* Same key is used for encryption and decryption.
* key must be distributed securely.

RSA:-

* Each user has a public-private key pair.
* Public key encrypts or wraps a key.
* Private key decrypts it.
* Digital certificates verify public-key identify.

Performance:-

AES 256-GCM efficiently encrypts 1GB

RSA cannot directly encrypt large files. For RSA-2048 with SHA-256 OAEP:-

$$\text{Maximum} = 256 - 2(32) - 2$$

190 bytes

Therefore, RSA is suitable for encryption of small keys.

Scalability:-

For 200 clinics communicating pairwise using symmetric encryption:-

$$N = \frac{200(199)}{2}$$

19,900 keys

For clinic-to-server communication, 200 shared keys are required.

* 200 clinic keypairs.
* 1 server keypair.
* Total = 201 keypairs

(3) b) Hybrid Encryption workflow

Hybrid encryption combines AES and RSA to provide confidentiality and secure key distribution.

(i) Sender side steps

* Certificate validation :- Verify the sender's digital certificate, CA signature, identify and validity.
* DEK generation :- Generate a random 256-bit AES Data Encryption key and unique 96-bit nonce.
* AES encryption :- Encrypt the MRI archive using AES-256-GCM with associated data (AAD).
* RSA key wrapping :- Encrypt the DEK using the server RSA public key with RSA-OAEP.
* Packageing :- Send ciphertext, encrypted DEK, nonce, authentication tag and required metadata.

Encryption :-

$$(C, T) = \text{AES-GCM}(DEK, N, P, AAD)$$

$$E = \text{RSA-OAEP}(\text{Publickey}, DEK)$$

(ii) Receiver - side steps

* Receive the encrypted package.
* Use the RSA private key to recover the DEK.
* Use the DEK and nonce to decrypt the MRI archeive.
* Verify the AES-GCM authentication tag.
* Accept the original data only if verification succeeds.

(iii) Secure key storage :-

Measure	Purpose
HSM/KMS	Secure key storage and management.
Non exportable keys	Prevent private key extraction
Role based access	Restrict unauthorized access
MFA	Strengthen authentication
Audit logging	Record key usage.
Backup and recovery	Recover keys during disasters.

D. Key Rotation

* Generate a new DEK for every file.
* Periodically rotate the KEK (Key Encryption key).
* Re-wrap existing DEKs under the new KEK when required.
* Retire old keys securely.
* Perform emergency rotation after suspected compromise.

4) RSA Algorithm

Given :- $P = 11, q = 19, e = 7, M = 5$

A) Calculate Modulus

$$n = P * q = 11 * 19$$

$$\boxed{n = 209}$$

(B) Calculate Euler's totient :-

$$\phi(n) = (p-1)(q-1)$$

$$= 10 \times 18$$

$$\phi(n) = 180$$

(C) Verify Public Exponent :-

$$\gcd(7, 180) = 1$$

Therefore, e is valid.

(D) Find Private Exponent using Extended Euclidean Algorithm :-

$$7d \equiv 1 \pmod{180}$$

$$180 = 25(7) + 5$$

$$7 = (1)(5) + 2$$

$$5 = 2(2) + 1$$

$$1 = 5 - 2(2)$$

$$= 5 - 2(7 - 5)$$

$$= 3(5) - 2(7)$$

$$3(180 - 25(7)) - 2(7)$$

$$= 3(180) - 77(7)$$

$$d = -77 \bmod 180$$

$$\boxed{d = 103}$$

E) Public and private keys:-

Public key – (7, 209)

Private key – (103, 209)

F) RSA Encryption:-

$C = M^e \bmod n$

$C = 5^7 \bmod 209$

$\boxed{C = 168}$

G) RSA Decryption using CRT:-

$M = c^d \bmod n$

$M = 168^{103} \bmod 209$

$\boxed{M = 5}$

The original message is successfully recovered.

H) Why Real systems Need Larger parameters:-

* Factoring attacks:- Small moduli can easily be factored. RSA-2048 or RSA-3072 provides stronger security.
* Determinism:- Textbook RSA produces identical ciphertext for identical plaintext. OAEP adds random padding.
* Malleability:- Textbook RSA allows mathematical manipulation of ciphertext.
* Small-message attacks:- Small messages can be vulnerable without secure padding.

I) RSA-OAEP and RSA-PSS

RSA-OAEP - Secure randomized padding for encryption

RSA-PSS- Secure randomized padding for digital signatures.

5) Classical and modern Encryption Techniques:-

Assume the plaintext:- TRANSFFRFUNDS

A) Caesar cipher

Given shift = 3

Formula:- $C = (P+3) \bmod 26$
↳ key

Ciphertext:- WUDQVI HUI XQGV

Decryption formula:- $P(C-3) \bmod 26$
↳ key

Recovered plaintext:- TRANSFERFUNDS

B) Playfair Cipher

Key= ~~MONARCHY~~ MONARCHY

Key matrix:-

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

Plaintext pairs:- TR AN SF ER FU ND SX

Ciphertext pairs :-ZD RA PI KM EV RY XA

Encrypted text :- ZDRAPIKMEVRYXA

Decryption uses reverse row and column movements.

Recovered plaintext:-TRANSFERFUNDS.

c) Vignere Cipher:-

Key= BANK

Repeated key= BANKBANKBANKB

Formula:- $C_i = (P_i + K_i) \bmod 26$

Plaintext	T	R	A	N	S	F	E	R	F	U	N	D	S	
Key	B	A	N	K	B	A	N	K	B	A	N	K	B	
Ciphertext	U	R	N	X	T	F	R	B	G	U	A	N	T	

Encrypted Text :- URNXTFRBGUANT

Decryption formula:-

$P_i = (C_i - K_i) \bmod 26$

Recovered plaintext:-TRANSFERFUNDS.

D) Comparison of classical ciphers :-

Parameter	Caesor	Playfair	Vigenere
Key	Fixed shift	Matrix keyword	Repeating keyword.
Method	single-letter	letter pairs	Multiple letters
Complexity	low	medium	medium
Security	Very weak	weak	weak
Attack	Brute force	Frequency analysis	Kasiski analysis.

E) Modern Encryption - AES-256-GCM

AES-256-GCM is selected to protect financial information because it provides confidentiality and integrity

Encryption steps :-

* Generates a random 256 bit AES key
* Generate a unique 96 bit once
* provide a Financial message as plaintext.
* Encrypt using AES-256-GCM with associated data.
* Generate ciphertext and authentication tag.
* Transmit ciphertext, nonce, tag and required metadata securely.

Formula :- $(C,T) = AES\text{-}GCM(K,N,P,AAD)$

Decryption :-

* Receiver obtains the encrypted data
* Uses the same secret key and nonce.
* Decrypts the ciphertext.
* Verifies the authentication tag.
* Accepts the message only if authentication succeeds.

$$P = AES\text{-}GCM^{-1}(K, N, C, T, AAD)$$

Recovered plaintext :- TRANSFER FUNDS

F) Classical vs modern Encryption :-

Parameter	Classical ciphers	AES-256 GCM
Key size	Small	256 bits
Security	Weak	strong
confidentiality	Limited	strong
Integrity	Not provided	Provided
Practical usage	Education	Modern communication

G) Suitable techniques for Financial Data :-

AES-256-GCM is the most suitable among the discussed techniqe because :-

* Provides strong confidentiality
* Uses a 256-bit key.
* Detects unauthorized modification.
* Provides authentication through an authentication tag
* Efficiently encrypts large financial records.